

```
In [ ]: print("Hello, Machine Learning!")
        print("I am learning Python using Google Colab.")
```

Hello, Machine Learning!
I am learning Python using Google Colab.

```
In [ ]: name = "Ahmed"
        age = 20
        height = 1.75
        is_student = True
        print(name)
        print(age)
        print(height)
        print(is_student)
        print(type(name))
        print(type(age))
        print(type(height))
        print(type(is_student))
```

Ahmed
20
1.75
True
<class 'str'>
<class 'int'>
<class 'float'>
<class 'bool'>

```
In [ ]: a = 10
        b = 3
        print("Addition:", a + b)
        print("Subtraction:", a - b)
        print("Multiplication:", a * b)
        print("Division:", a / b)
        print("Floor division:", a // b)
        print("Remainder:", a % b)
        print("Power:", a ** b)
```

Addition: 13
Subtraction: 7
Multiplication: 30
Division: 3.3333333333333335
Floor division: 3
Remainder: 1
Power: 1000

```
In [ ]: name = input("Enter your name: ")
        age = int(input("Enter your age: "))
        print("Hello", name)
        print("Next year you will be", age + 1)
```

Enter your name: Ahmed
Enter your age: 26
Hello Ahmed
Next year you will be 27

```
In [1]: mark = float(input("Enter your mark: "))

        if mark >= 60:
            print("Pass")
```

```

else:
    print("Fail")

if mark >= 90:
    print("Grade: A")
elif mark >= 80:
    print("Grade: B")
elif mark >= 70:
    print("Grade: C")
elif mark >= 60:
    print("Grade: D")
else:
    print("Grade: F")

```

Enter your mark: 99

Pass

Grade: A

```

In [ ]: marks = [75, 82, 91, 68, 88]
        print("Marks:", marks)
        print("First mark:", marks[0])
        print("Third mark:", marks[2])
        print("Number of marks:", len(marks))
        print("Total:", sum(marks))
        print("Highest:", max(marks))
        print("Lowest:", min(marks))
        print("Average:", sum(marks) / len(marks))

```

Marks: [75, 82, 91, 68, 88]

First mark: 75

Third mark: 91

Number of marks: 5

Total: 404

Highest: 91

Lowest: 68

Average: 80.8

```

In [ ]: temperature = [25.4, 26.1, 27.3, 28.0, 26.8]
        print("Number of observations:", len(temperature))
        print("Average temperature:", sum(temperature) / len(temperature))
        print("Minimum temperature:", min(temperature))
        print("Maximum temperature:", max(temperature))

```

Number of observations: 5

Average temperature: 26.72

Minimum temperature: 25.4

Maximum temperature: 28.0

```

In [3]: study_hours = [1, 2, 3, 4, 5, 6, 7]
        marks = [50, 55, 60, 65, 72, 80, 88]

        for i in range(len(study_hours)):
            print("Study hours:", study_hours[i], "Marks:", marks[i])

```

Study hours: 1 Marks: 50

Study hours: 2 Marks: 55

Study hours: 3 Marks: 60

Study hours: 4 Marks: 65

Study hours: 5 Marks: 72

Study hours: 6 Marks: 80

Study hours: 7 Marks: 88

```
In [4]: study_hours = [1, 2, 3, 4, 5, 6, 7]
marks = [50, 55, 60, 65, 72, 80, 88]

for mark in marks:
    print(mark)

for mark in marks:
    if mark >= 60:
        print(mark, "Pass")
    else:
        print(mark, "Fail")

for i in range(len(marks)):
    if marks[i] >= 60:
        result = "Pass"
    else:
        result = "Fail"
    print("Study hours:", study_hours[i],
          "Mark:", marks[i], "Result:", result)
```

```
50
55
60
65
72
80
88
50 Fail
55 Fail
60 Pass
65 Pass
72 Pass
80 Pass
88 Pass
Study hours: 1 Mark: 50 Result: Fail
Study hours: 2 Mark: 55 Result: Fail
Study hours: 3 Mark: 60 Result: Pass
Study hours: 4 Mark: 65 Result: Pass
Study hours: 5 Mark: 72 Result: Pass
Study hours: 6 Mark: 80 Result: Pass
Study hours: 7 Mark: 88 Result: Pass
```

```
In [5]: study_hours = [1, 2, 3, 4, 5, 6, 7]
marks = [50, 55, 60, 65, 72, 80, 88]

n = len(marks)
print("Number of samples:", n)

total_hours = 0
for h in study_hours:
    total_hours = total_hours + h
print("Average study hours:", total_hours / n)

total_marks = 0
for m in marks:
    total_marks = total_marks + m
print("Average marks:", total_marks / n)

highest = marks[0]
lowest = marks[0]
```

```

for m in marks:
    if m > highest:
        highest = m
    if m < lowest:
        lowest = m
print("Highest mark:", highest)
print("Lowest mark:", lowest)

for i in range(n):
    print("Study hours:", study_hours[i], "Mark:", marks[i])

for i in range(n):
    if marks[i] >= 60:
        print("Study hours:", study_hours[i], "Mark:", marks[i], "Pass")
    else:
        print("Study hours:", study_hours[i], "Mark:", marks[i], "Fail")

count = 0
for m in marks:
    if m >= 60:
        count = count + 1
print("Number of passes:", count)

```

Number of samples: 7
 Average study hours: 4.0
 Average marks: 67.14285714285714
 Highest mark: 88
 Lowest mark: 50
 Study hours: 1 Mark: 50
 Study hours: 2 Mark: 55
 Study hours: 3 Mark: 60
 Study hours: 4 Mark: 65
 Study hours: 5 Mark: 72
 Study hours: 6 Mark: 80
 Study hours: 7 Mark: 88
 Study hours: 1 Mark: 50 Fail
 Study hours: 2 Mark: 55 Fail
 Study hours: 3 Mark: 60 Pass
 Study hours: 4 Mark: 65 Pass
 Study hours: 5 Mark: 72 Pass
 Study hours: 6 Mark: 80 Pass
 Study hours: 7 Mark: 88 Pass
 Number of passes: 5